

Kartik Narayan

LinkedIn | Github | Google Scholar
<https://kartik-3004.github.io/portfolio/>

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Summary

I am a 3rd year (final year) Ph.D. student in the Computer Science Department at Johns Hopkins University. My current research interests lie in multimodal LLMs, agents, reasoning and video generation.

Graduation Date: December 2026

Education

Johns Hopkins University

Ph.D. Computer Science, Advisor: Dr. Vishal M. Patel

Baltimore, MD

2023 - Present

Indian Institute of Technology Jodhpur

Bachelors in Computer Science and Engineering

Jodhpur, India

2019 - 2023

Publications

- DeepMMSearch-R1: Empowering Multimodal LLMs in Multimodal Web Search**
Kartik Narayan, Yang Xu, Tian Cao, Kavya Nerella, Vishal M. Patel, Navid Shiee, Peter Grasch, Chao Jia, Yinfei Yang, Zhe Gan
Under Review. [\[arXiv\]](#)
- FaceMoE: Mixture of Experts for Low-resolution Face Recognition**
Kartik Narayan, Vishal M. Patel
European Conference on Computer Vision 2026 (ECCV 2026) [\[paper\]](#)
- FaceXFormer: A Unified Transformer for Facial Analysis**
Kartik Narayan, Vibashan VS*, Rama Chellappa, Vishal M. Patel*
Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV 2025)
- SegFace: Face Segmentation of Long-Tail Classes**
Kartik Narayan, Vibashan VS, Vishal M. Patel
Association for the Advancement of Artificial Intelligence (AAAI 2025)
- DF-Platter: Multi-Face Heterogeneous Deepfake Dataset**
Kartik Narayan, Harsh Agarwal, Kartik Thakral, Surbhi Mittal, Mayank Vatsa, Richa Singh
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2023)
- RestoreVAR: Visual Autoregressive Generation for All-in-One Image Restoration**
Sudarshan Rajagopalan, Kartik Narayan, Vishal M. Patel
International Conference on Learning Representations 2026 (ICLR 2026) [\[arXiv\]](#)
- Silhouette-based Gait Foundation Model**
Dingqiang Ye, Chao Fan, Kartik Narayan, Bingzhe Wu, Chengwen Luo, Jianqiang Li, Vishal M. Patel
European Conference on Computer Vision 2026 (ECCV 2026) [\[arXiv\]](#)
- FaceXBench: Evaluating Multimodal LLMs on Face Understanding**
Kartik Narayan, Vibashan VS, Vishal M. Patel
TBIOM [\[arXiv\]](#)
- PETALface: Parameter Efficient Transfer Learning for Low-resolution Face Recognition**
Kartik Narayan, Nithin Gopalakrishnan Nair, Jennifer Xu, Rama Chellappa, Vishal M. Patel
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2025) (Oral)
- Training-Free Stylized Abstraction**
Aiman Rahman, Kartik Narayan*, Vishal M. Patel*
Under Review. [\[arXiv\]](#)

11. **TransFIRA: Transfer Learning for Face Image Recognizability Assessment**
*Allen Tu, **Kartik Narayan**, Joshua Gleason, Jennifer Xu, Matthew Meyn, Tom Goldstein, Vishal M. Patel*
 2026 20th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2026). [[arXiv](#)]
12. **Investigating Social Biases in Multimodal LLMs**
Malsha Perera, **Kartik Narayan***, Vishal M. Patel*
 2025 19th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2025).
13. **Improved Representation Learning for Unconstrained Face Recognition**
Nithin Gopalakrishnan Nair, **Kartik Narayan***, Maitreya Suin, Ram Prabhakar, Jennifer Xu, Soraya Stevens, Joshua Gleason, Nathan Shindman, Rama Chellappa, Vishal M. Patel*
 2025 19th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2025).
14. **INFER: Implicit Neural Features for Exposing Realism**
*Dhananjaya Jayasundara, **Kartik Narayan**, Vishal M. Patel*
 Under Review. [[paper](#)]
15. **Hyp-OC: Hyperbolic One Class Classification for Face Anti-Spoofing**
***Kartik Narayan**, Vishal M. Patel*
 2024 18th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2024)
16. **DeePhyNet: Towards Detecting Phylogeny in Deepfakes**
*Kartik Thakral, Harsh Agarwal, **Kartik Narayan**, Surbhi Mittal, Mayank Vatsa, Richa Singh*
 IEEE Transactions on Biometrics, Behavior, and Identity Science (T-BIOM)
17. **DeePhy: On Deepfake Phylogeny**
***Kartik Narayan**, Harsh Agarwal, Kartik Thakral, Surbhi Mittal, Mayank Vatsa, Richa Singh*
 2022 IEEE International Joint Conference on Biometrics (IJCB 2022)
18. **DeSI: Deepfake Source Identifier for Social Media**
***Kartik Narayan**, Harsh Agarwal, Surbhi Mittal, Kartik Thakral, Suman Kundu, Mayank Vatsa, Richa Singh*
 Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW 2022)
19. **Using Epidemic Modeling, Machine Learning & Control Feedback Strategy for Policy Management of COVID-19**
***Kartik Narayan**, Heena Rathore, Faycal Znidi*
 IEEE Access 10 (2022): 98244–98258
20. **Leveraging Ambient Sensing for the Estimation of Curiosity-Driven Human Crowd**
*Anirban Das, **Kartik Narayan**, Suchetana Chakraborty*
 2022 IEEE International Systems Conference (SysCon 2022)

Research Experience

Salesforce

Ph.D. Research Intern

Palo Alto, CA

Summer 2026

- Multimodal LLMs for deep-research
- Working on instilling self-verification behavior in MLLMs using agentic RL training for MLLMs
- Working on Video-DeepResearch evaluation and Benchmark.

Apple Inc.

Ph.D. Research Intern

Seattle, WA

Summer 2025 and Spring 2026

- Multimodal LLMs for knowledge-intensive and information seeking questions
- Equipping MLLMs with web-search agent and enabling MLLM reasoning for web-search calls.
- Incentivizing MLLM for self-correction and self-verification after initial response.

Johns Hopkins University

Research Assistant, Vision and Image Understanding Lab advised by Dr. Vishal M. Patel

Baltimore, Maryland

2023 - Present

- Leading research efforts for the IARPA BRIAR grant, focusing on face recognition in extreme scenarios.

- Worked on designing a multi-task model for multiple face analysis tasks.
- Evaluated Multimodal LLMs on face understanding, and explored future directions such as SFT and tools use.
- Currently working on video diffusion models for physics-aware video generation.
- Tackling fundamental issues in T2V models such as text-rich video generation.

Indian Institute of Technology Jodhpur

Student Researcher, IAB Lab advised by Prof. Richa Singh and Prof. Mayank Vatsa

Jodhpur, India

2022 - 2023

- Worked on deepfake video generation, with a special emphasize on multi-face, low-resolution, and occluded videos.
- Introduced the concept of phylogeny (evolution) in deepfake video generation by sequentially swapping multiple faces.
- Developed the DeSI algorithm, capable of detecting deepfake videos on the Twitter platform and predicting their spread.

Academic Services

- **Invited Reviewer:** CVPR, ICCV, ECCV, AAAI, NeurIPS, ICLR, ICML, TPAMI, WaCV, IJCB, TIFS, TBIOM, IJCV, PR

Skills Summary

- **Programming:** Python, C/C++
- **ML/DL:** PyTorch, TensorFlow, Keras, OpenCV, Sklearn, Numpy, Pandas, Matplotlib
- **Development:** HTML, CSS, React.js, Node.js, Bootstrap, Firebase, MongoDB, ReactNative
- **Tools:** Docker, Kubernetes, GIT, AWS

Teaching Experience

Teaching Assistantship: Assisted in teaching by conducting weekly lab sessions, holding special doubt-clearing sessions, preparing quizzes, and grading assignments for the following courses:

- Deep Learning [Spring 2023]
- Introduction to Machine Learning [Fall 2022]
- Pattern Recognition and Machine Learning [Spring 2022]

Co-curricular Activities

- **Internship Head, CSE at the placement cell of IIT Jodhpur** [2021 - 2022]
Coordinated with companies to invite them for campus internships and placements.
- **Head of technical events in Prometeo, IIT Jodhpur** [2021 - 2022]
Led a team of 50+ students to organize technical competitions, attracting over 1,500 participants from across India.
- **Secretary of PHEME, The IITJ Newsletter Club** [2020 - 2021]
Managed a club of 80+ members responsible for creating newsletters, reports, conducting surveys, etc.
- **Student Guide at Student Wellbeing Committee, IITJ** [2020 - 2023]
Mentored 10 freshmen from diverse backgrounds to support their transition into college, guiding them in both academic and non-academic pursuits.